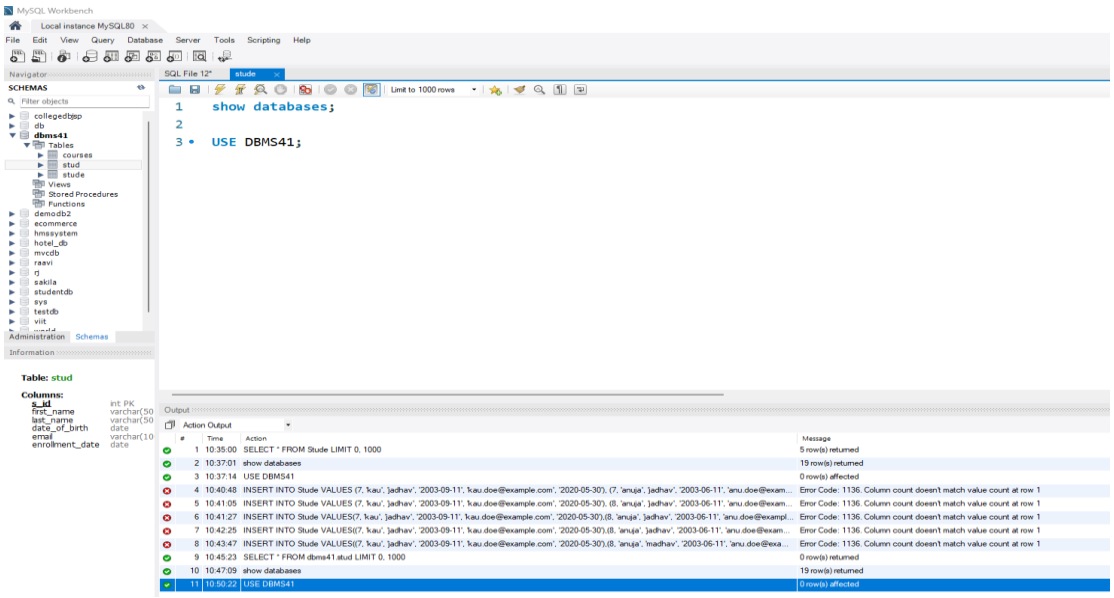
SQL Queries and Joins

NIKITA SANJAY MANE (41)

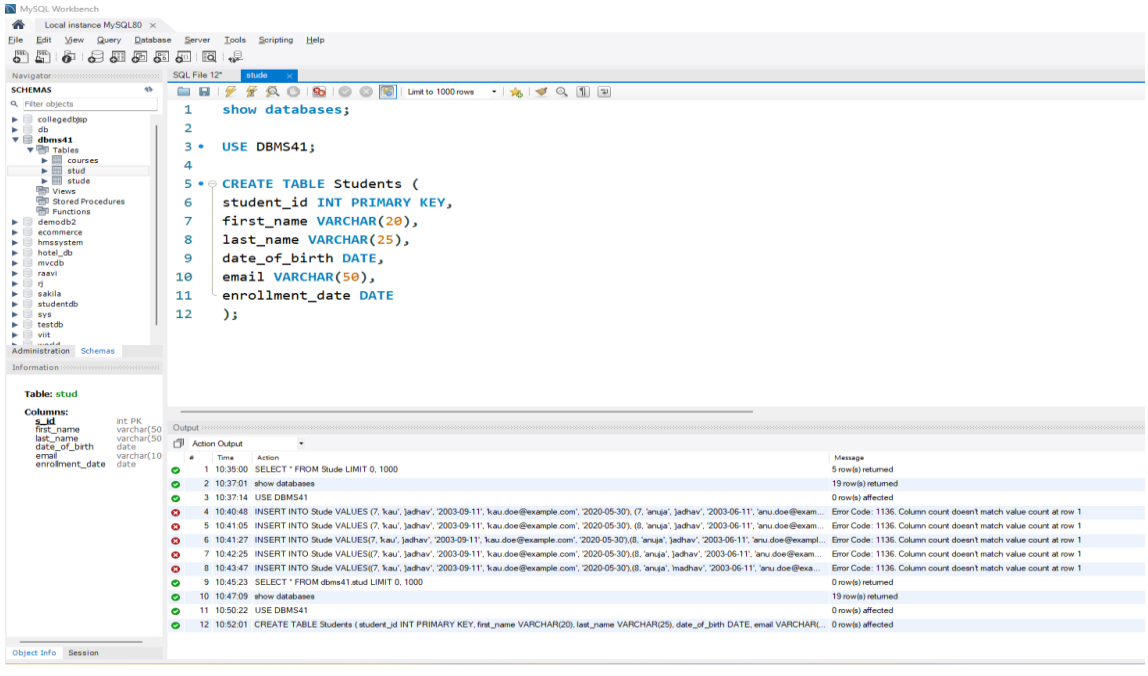
1. Creating a Database

```
CREATE DATABASE DBMS41;
```

2. Using the Database



3. Creating Tables



4. Inserting Data into Tables

```
15
16  #Inserting Data into Tables
17  INSERT INTO Students VALUES (1, 'John', 'Doe', '2000-03-15', 'john.doe@example.com', '2020-09-01');
18
19  INSERT INTO Students VALUES(2, 'Jane', 'Smith', '1999-05-20', 'jane.smith@example.com', '2020-09-01'),
20  (3, 'Alice', 'Johnson', '2001-07-12', 'alice.johnson@example.com', '2021-01-15'),
21
22  INSERT INTO Students VALUES(4, 'Neha', 'mahamulkar', '2005-11-25', 'neha.johnson@example.com', '2021-02-16'),
23  (5, 'Kaumudhi', 'Sutar', '2004-09-09', 'Kau.johnson@example.com', '2021-03-17'),
24  (6, 'Ridhhi', 'Jadhav', '2005-03-19', 'Don.johnson@example.com', '2021-04-18');
```

Output

Action Output

#	Time	Action	Message
8	10:43:47	INSERT INTO Stude VALUES((7, 'kau', 'jadhav', '2003-09-11', 'kau.doe@example.com', '2020-05-30'),(8, 'anuja', 'madhav', '2003-06-11', 'anu.doe@ex...	Error Code: 1136. Column count doesn't match value count at row 1
9	10:45:23	SELECT * FROM dbms41.stud LIMIT 0, 1000	0 row(s) returned
10	10:47:09	show databases	19 row(s) returned
11	10:50:22	USE DBMS41	0 row(s) affected
12	10:52:01	CREATE TABLE Students (student_id INT PRIMARY KEY, first_name VARCHAR(20), last_name VARCHAR(25), date_of_birth DATE, email VARCHA...	0 row(s) affected
13	10:55:23	INSERT INTO Students VALUES (1, 'John', 'Doe', '2000-03-15', 'john.doe@example.com', '2020-09-01')	1 row(s) affected
14	10:55:28	INSERT INTO Students VALUES(2, 'Jane', 'Smith', '1999-05-20', 'jane.smith@example.com', '2020-09-01'), (3, 'Alice', 'Johnson', '2001-07-12', 'alice.john...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
15	11:01:16	INSERT INTO Students VALUES(4, 'Neha', 'mahamulkar', '2005-11-25', 'neha.johnson@example.com', '2021-02-16'), (5, 'Kaumudhi', 'Sutar', '2004-09-0...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

5. Retrieving Data (SELECT)

```
26  #Retrieving Data (SELECT)
27  select * from Students;
28
29  #Retrieve Specific Columns:
30  SELECT first_name, last_name, email FROM Students;
31
32  #Retrieve Students Enrolled After a Specific Date:
33  SELECT * FROM Students
34  WHERE enrollment_date > '2020-12-31';
35
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

student_id	first_name	last_name	date_of_birth	email	enrollment_date	phone_number
3	Alice	Johnson	2001-07-12	alice.johnson@example.com	2021-01-15	NULL
4	Neha	mahamulkar	2005-11-25	neha.johnson@example.com	2021-02-16	NULL
5	Kaumudhi	Sutar	2004-09-09	Kau.johnson@example.com	2021-03-17	NULL
6	Ridhhi	Jadhav	2005-03-19	Don.johnson@example.com	2021-04-18	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Students 4

Output

Action Output

#	Time	Action	Message
1	13:38:29	SELECT F1.first_name AS Faculty_Name, F2.first_name AS Manager_Name FROM Faculty F1 LEFT JOIN Faculty F2 ON F1.manager_id = F2.faculty_id...	0 row(s) returned
2	14:05:10	UPDATE Students SET email = 'john.doe2025@example.com' WHERE student_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
3	14:07:23	select * from Students LIMIT 0, 1000	5 row(s) returned
4	14:08:43	SELECT first_name, last_name, email FROM Students LIMIT 0, 1000	5 row(s) returned
5	14:10:23	SELECT * FROM Students WHERE enrollment_date > '2020-12-31' LIMIT 0, 1000	4 row(s) returned

i.

```
25
26    #Retrieving Data (SELECT)
27 •  select * from Students;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

student_id	first_name	last_name	date_of_birth	email	enrollment_date	phone_number
1	John	Doe	2000-03-15	john.doe2025@example.com	2020-09-01	NULL
3	Alice	Johnson	2001-07-12	alice.johnson@example.com	2021-01-15	NULL
4	Neha	mahamulkar	2005-11-25	neha.johnson@example.com	2021-02-16	NULL
5	Kaumudhi	Sutar	2004-09-09	Kau.johnson@example.com	2021-03-17	NULL
6	Ridhhi	Jadhav	2005-03-19	Don.johnson@example.com	2021-04-18	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Students 2

Output

Action Output

#	Time	Action	Message
1	13:38:29	SELECT F1.first_name AS Faculty_Name, F2.first_name AS Manager_Name FROM Faculty F1 LEFT JOIN Faculty F2 ON F1.manager_id = F2.faculty_id...	0 row(s) returned
2	14:05:10	UPDATE Students SET email = 'john.doe2025@example.com' WHERE student_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
3	14:07:23	select * from Students LIMIT 0, 1000	5 row(s) returned

ii.

```
29    #Retrieve Specific Columns:
30 •  SELECT first_name, last_name, email FROM Students;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	email
John	Doe	john.doe2025@example.com
Alice	Johnson	alice.johnson@example.com
Neha	mahamulkar	neha.johnson@example.com
Kaumudhi	Sutar	Kau.johnson@example.com
Ridhhi	Jadhav	Don.johnson@example.com

Students 3

Output

Action Output

#	Time	Action	Message
1	13:38:29	SELECT F1.first_name AS Faculty_Name, F2.first_name AS Manager_Name FROM Faculty F1 LEFT JOIN Faculty F2 ON F1.manager_id = F2.faculty_id...	0 row(s) returned
2	14:05:10	UPDATE Students SET email = 'john.doe2025@example.com' WHERE student_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
3	14:07:23	select * from Students LIMIT 0, 1000	5 row(s) returned
4	14:08:43	SELECT first_name, last_name, email FROM Students LIMIT 0, 1000	5 row(s) returned

iii.

```
32    #Retrieve Students Enrolled After a Specific Date:
33 •  SELECT * FROM Students
34    WHERE enrollment_date > '2020-12-31';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

student_id	first_name	last_name	date_of_birth	email	enrollment_date	phone_number
3	Alice	Johnson	2001-07-12	alice.johnson@example.com	2021-01-15	NULL
4	Neha	mahamulkar	2005-11-25	neha.johnson@example.com	2021-02-16	NULL
5	Kaumudhi	Sutar	2004-09-09	Kau.johnson@example.com	2021-03-17	NULL
6	Ridhhi	Jadhav	2005-03-19	Don.johnson@example.com	2021-04-18	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Students 4

Output

Action Output

#	Time	Action	Message
1	13:38:29	SELECT F1.first_name AS Faculty_Name, F2.first_name AS Manager_Name FROM Faculty F1 LEFT JOIN Faculty F2 ON F1.manager_id = F2.faculty_id...	0 row(s) returned
2	14:05:10	UPDATE Students SET email = 'john.doe2025@example.com' WHERE student_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
3	14:07:23	select * from Students LIMIT 0, 1000	5 row(s) returned
4	14:08:43	SELECT first_name, last_name, email FROM Students LIMIT 0, 1000	5 row(s) returned
5	14:10:23	SELECT * FROM Students WHERE enrollment_date > '2020-12-31' LIMIT 0, 1000	4 row(s) returned

6. Updating Data (UPDATE)

```
31
32 •  UPDATE Students
33    SET email = 'john.doe2025@example.com'
34    WHERE student_id = 1;
```

Output

Action Output

#	Time	Action	Message
1	13:38:29	SELECT F1.first_name AS Faculty_Name, F2.first_name AS Manager_Name FROM Faculty F1 LEFT JOIN Faculty F2 ON F1.manager_id = F2.faculty_id...	0 row(s) returned
2	14:05:10	UPDATE Students SET email = 'john.doe2025@example.com' WHERE student_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

7. Deleting Data (DELETE)

```
32 •  DELETE FROM Students
33    WHERE student_id = 2;
34
```

Output

Action Output

#	Time	Action	Message
11	10:50:22	USE DBMS41	0 row(s) affected
12	10:52:01	CREATE TABLE Students (student_id INT PRIMARY KEY, first_name VARCHAR(20), last_name VARCHAR(25), date_of_birth DATE, email VARCHA...	0 row(s) affected
13	10:55:23	INSERT INTO Students VALUES (1, 'John', 'Doe', '2000-03-15', 'john.doe@example.com', '2020-09-01')	1 row(s) affected
14	10:55:28	INSERT INTO Students VALUES(2, 'Jane', 'Smith', '1999-05-20', 'jane.smith@example.com', '2020-09-01'), (3, 'Alice', 'Johnson', '2001-07-12', 'alice.john...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
15	11:01:16	INSERT INTO Students VALUES(4, 'Neha', 'mahamulkar', '2005-11-25', 'neha.johnson@example.com', '2021-02-16'), (5, 'Kaumudhi', 'Sutar', '2004-09-0...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
16	11:03:12	select * from Students LIMIT 0, 1000	6 row(s) returned
17	11:04:36	SELECT * FROM Students WHERE enrollment_date > '2020-12-31' LIMIT 0, 1000	4 row(s) returned
18	11:07:26	DELETE FROM Students WHERE student_id = 2	1 row(s) affected

8. Adding a Column to a Table

```
34
35 •  ALTER TABLE Students
36    ADD phone_number VARCHAR(15);
```

Output

Action Output

#	Time	Action	Message
12	10:52:01	CREATE TABLE Students (student_id INT PRIMARY KEY, first_name VARCHAR(20), last_name VARCHAR(25), date_of_birth DATE, email VARCHA...	0 row(s) affected
13	10:55:23	INSERT INTO Students VALUES (1, 'John', 'Doe', '2000-03-15', 'john.doe@example.com', '2020-09-01')	1 row(s) affected
14	10:55:28	INSERT INTO Students VALUES(2, 'Jane', 'Smith', '1999-05-20', 'jane.smith@example.com', '2020-09-01'), (3, 'Alice', 'Johnson', '2001-07-12', 'alice.john...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
15	11:01:16	INSERT INTO Students VALUES(4, 'Neha', 'mahamulkar', '2005-11-25', 'neha.johnson@example.com', '2021-02-16'), (5, 'Kaumudhi', 'Sutar', '2004-09-0...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
16	11:03:12	select * from Students LIMIT 0, 1000	6 row(s) returned
17	11:04:36	SELECT * FROM Students WHERE enrollment_date > '2020-12-31' LIMIT 0, 1000	4 row(s) returned
18	11:07:26	DELETE FROM Students WHERE student_id = 2	1 row(s) affected
19	11:08:51	ALTER TABLE Students ADD phone_number VARCHAR(15)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

9. Dropping a Table

```
37
38 •  DROP TABLE stude;
```

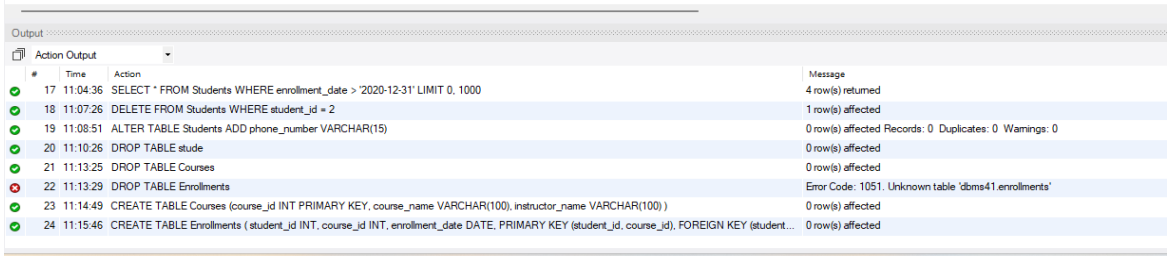
Output

Action Output

#	Time	Action	Message
13	10:55:23	INSERT INTO Students VALUES (1, 'John', 'Doe', '2000-03-15', 'john.doe@example.com', '2020-09-01')	1 row(s) affected
14	10:55:28	INSERT INTO Students VALUES(2, 'Jane', 'Smith', '1999-05-20', 'jane.smith@example.com', '2020-09-01'), (3, 'Alice', 'Johnson', '2001-07-12', 'alice.john...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
15	11:01:16	INSERT INTO Students VALUES(4, 'Neha', 'mahamulkar', '2005-11-25', 'neha.johnson@example.com', '2021-02-16'), (5, 'Kaumudhi', 'Sutar', '2004-09-0...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
16	11:03:12	select * from Students LIMIT 0, 1000	6 row(s) returned
17	11:04:36	SELECT * FROM Students WHERE enrollment_date > '2020-12-31' LIMIT 0, 1000	4 row(s) returned
18	11:07:26	DELETE FROM Students WHERE student_id = 2	1 row(s) affected
19	11:08:51	ALTER TABLE Students ADD phone_number VARCHAR(15)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
20	11:10:26	DROP TABLE stude	0 row(s) affected

10. Creating Relationships (Foreign Keys)

```
CREATE TABLE Enrollments (
student_id INT,
course_id INT,
enrollment_date DATE,
PRIMARY KEY (student_id, course_id),
FOREIGN KEY (student_id) REFERENCES Students(student_id),
FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);
```



Output

Action Output

#	Time	Action	Message
17	11:04:36	SELECT * FROM Students WHERE enrollment_date > '2020-12-31' LIMIT 0, 1000	4 row(s) returned
18	11:07:26	DELETE FROM Students WHERE student_id = 2	1 row(s) affected
19	11:08:51	ALTER TABLE Students ADD phone_number VARCHAR(15)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
20	11:10:26	DROP TABLE stude	0 row(s) affected
21	11:13:25	DROP TABLE Courses	0 row(s) affected
22	11:13:29	DROP TABLE Enrollments	Error Code: 1051. Unknown table 'dbms41.enrollments'
23	11:14:49	CREATE TABLE Courses (course_id INT PRIMARY KEY, course_name VARCHAR(100), instructor_name VARCHAR(100))	0 row(s) affected
24	11:15:46	CREATE TABLE Enrollments (student_id INT, course_id INT, enrollment_date DATE, PRIMARY KEY (student_id, course_id), FOREIGN KEY (student...	0 row(s) affected

11. Using JOINs (Combining Data from Multiple Tables)

```
SELECT Students.first_name, Students.last_name, Courses.course_name
FROM Students
INNER JOIN Enrollments ON Students.student_id = Enrollments.student_id
INNER JOIN Courses ON Enrollments.course_id = Courses.course_id;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	course_name

Result 4

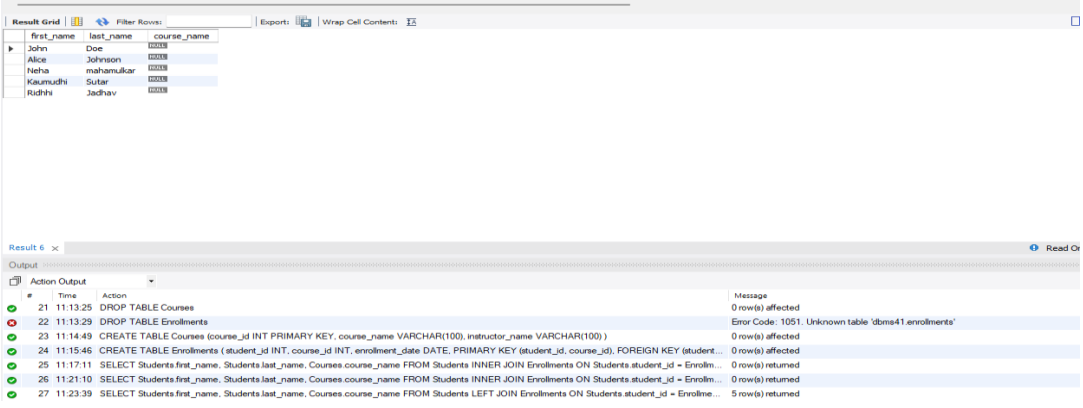
Output

Action Output

#	Time	Action	Message
19	11:08:51	ALTER TABLE Students ADD phone_number VARCHAR(15)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
20	11:10:26	DROP TABLE stude	0 row(s) affected
21	11:13:25	DROP TABLE Courses	0 row(s) affected
22	11:13:29	DROP TABLE Enrollments	Error Code: 1051. Unknown table 'dbms41.enrollments'
23	11:14:49	CREATE TABLE Courses (course_id INT PRIMARY KEY, course_name VARCHAR(100), instructor_name VARCHAR(100))	0 row(s) affected
24	11:15:46	CREATE TABLE Enrollments (student_id INT, course_id INT, enrollment_date DATE, PRIMARY KEY (student_id, course_id), FOREIGN KEY (student...	0 row(s) affected
25	11:17:11	SELECT Students.first_name, Students.last_name, Courses.course_name FROM Students INNER JOIN Enrollments ON Students.student_id = Enrollm...	0 row(s) returned

i. INNER JOIN

```
SELECT Students.first_name, Students.last_name, Courses.course_name
FROM Students
LEFT JOIN Enrollments ON Students.student_id = Enrollments.student_id
LEFT JOIN Courses ON Enrollments.course_id = Courses.course_id;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	course_name
John	Doe	NULL
Alice	Johnson	NULL
Neha	mahamulkar	NULL
Kaumudhi	Sutar	NULL
Ridhhi	Jadhav	NULL

Result 6

Output

Action Output

#	Time	Action	Message
21	11:13:25	DROP TABLE Courses	0 row(s) affected
22	11:13:29	DROP TABLE Enrollments	Error Code: 1051. Unknown table 'dbms41.enrollments'
23	11:14:49	CREATE TABLE Courses (course_id INT PRIMARY KEY, course_name VARCHAR(100), instructor_name VARCHAR(100))	0 row(s) affected
24	11:15:46	CREATE TABLE Enrollments (student_id INT, course_id INT, enrollment_date DATE, PRIMARY KEY (student_id, course_id), FOREIGN KEY (student...	0 row(s) affected
25	11:17:11	SELECT Students.first_name, Students.last_name, Courses.course_name FROM Students INNER JOIN Enrollments ON Students.student_id = Enrollm...	0 row(s) returned
26	11:21:10	SELECT Students.first_name, Students.last_name, Courses.course_name FROM Students INNER JOIN Enrollments ON Students.student_id = Enrollm...	0 row(s) returned
27	11:23:39	SELECT Students.first_name, Students.last_name, Courses.course_name FROM Students LEFT JOIN Enrollments ON Students.student_id = Enrollme...	5 row(s) returned

ii. LEFT JOIN (or LEFT OUTER JOIN)

```
62
63 •  SELECT Students.first_name, Students.last_name, Courses.course_name
64    FROM Students
65    LEFT JOIN Enrollments ON Students.student_id = Enrollments.student_id
66    LEFT JOIN Courses ON Enrollments.course_id = Courses.course_id;
67
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	course_name
John	Doe	NULL
Alice	Johnson	NULL
Neha	mahamulkar	NULL
Kaumudhi	Sutar	NULL
Ridhhi	Jadhav	NULL

Result 7

Output

Action Output

#	Time	Action	Message
27	11:23:39	SELECT Students.first_name, Students.last_name, Courses.course_name FROM Students LEFT JOIN Enrollments ON Students.student_id = Enrollme...	5 row(s) returned
28	11:29:16	SELECT Students.first_name, Students.last_name, Courses.course_name FROM Students LEFT JOIN Enrollments ON Students.student_id = Enrollme...	5 row(s) returned

iii. RIGHT JOIN (or RIGHT OUTER JOIN)

```
67
68 •  SELECT Students.first_name, Students.last_name, Courses.course_name
69    FROM Students
70    RIGHT JOIN Enrollments ON Students.student_id = Enrollments.student_id
71    RIGHT JOIN Courses ON Enrollments.course_id = Courses.course_id;
72
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	course_name

Result 8

Output

Action Output

#	Time	Action	Message
28	11:29:16	SELECT Students.first_name, Students.last_name, Courses.course_name FROM Students LEFT JOIN Enrollments ON Students.student_id = Enrollme...	5 row(s) returned
29	11:30:45	SELECT Students.first_name, Students.last_name, Courses.course_name FROM Students RIGHT JOIN Enrollments ON Students.student_id = Enrollm...	0 row(s) returned

```
90
91 • CREATE TABLE Faculty (
92   faculty_id INT PRIMARY KEY,
93   first_name VARCHAR(50),
94   last_name VARCHAR(50),
95   manager_id INT
96   );
97
98 •  SELECT F1.first_name AS Faculty_Name, F2.first_name AS Manager_Name
99   FROM Faculty F1
100  LEFT JOIN Faculty F2 ON F1.manager_id = F2.faculty_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Faculty_Name	Manager_Name

Result 5

Output

Action Output

#	Time	Action	Message
1	13:38:29	SELECT F1.first_name AS Faculty_Name, F2.first_name AS Manager_Name FROM Faculty F1 LEFT JOIN Faculty F2 ON F1.manager_id = F2.faculty_id...	0 row(s) returned
2	14:05:10	UPDATE Students SET email = 'john.doe2025@example.com' WHERE student_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
3	14:07:23	select * from Students LIMIT 0, 1000	5 row(s) returned
4	14:08:43	SELECT first_name, last_name, email FROM Students LIMIT 0, 1000	5 row(s) returned
5	14:10:23	SELECT * FROM Students WHERE enrollment_date > '2020-12-31' LIMIT 0, 1000	4 row(s) returned
6	15:04:53	SELECT F1.first_name AS Faculty_Name, F2.first_name AS Manager_Name FROM Faculty F1 LEFT JOIN Faculty F2 ON F1.manager_id = F2.faculty_id...	0 row(s) returned

iv. FULL JOIN (or FULL OUTER JOIN)

```
SELECT Students.first_name, Students.last_name, Courses.course_name
FROM Students
FULL JOIN Enrollments ON Students.student_id = Enrollments.student_id
FULL JOIN Courses ON Enrollments.course_id = Courses.course_id;
```

v. CROSS JOIN

```
78
79 •  SELECT Students.first_name, Students.last_name, Courses.course_name
80   FROM Students
81   CROSS JOIN Courses;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	course_name

Result 9

Output

Action Output

#	Time	Action	Message
31	11:33:05	SELECT Students.first_name, Students.last_name, Courses.course_name FROM Students FULL JOIN Enrollments ON Students.student_id = Enrollme...	Error Code: 1054. Unknown column 'Students.first_name' in 'field list'
32	11:34:18	SELECT Students.first_name, Students.last_name, Courses.course_name FROM Students CROSS JOIN Courses LIMIT 0, 1000	0 row(s) returned

vi. SELF JOIN